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Signal Decomposition in Stellar Light Curves

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Abstract

We demonstrate the RMxAA journal template compiled from markdown using the Inkwell extension. Fourier analysis of periodic signals is presented as a computational example, with figures generated from Python code blocks and embedded directly in the output. This literate programming approach produces reproducible, submission-ready manuscripts.

Resumen

Demostramos la plantilla de la revista RMxAA compilada desde markdown usando la extension Inkwell. Se presenta el analisis de Fourier de senales periodicas como ejemplo computacional, con figuras generadas desde bloques de codigo Python e integradas directamente en la salida.

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1. INTRODUCTION

Stellar light curve analysis often requires decomposing periodic signals into frequency components. The Fourier series provides the mathematical framework for this decomposition (**?**), and modern computational tools allow rapid visualization of convergence properties.

This document demonstrates the RMxAA journal template compiled from markdown through Pandoc () and LaTeX. Code blocks execute in place, producing figures that appear in the compiled PDF.

2. FOURIER ANALYSIS

The partial sum of a square wave's Fourier series is

$$f_n(x) = \sum_{k=1}^{n} \frac{4}{(2k-1)\pi} \sin\big((2k-1)x\big). \tag{1}$$

Figure 1 shows the convergence for increasing n.

```
{python file=".inkwell/scripts/sine_plot.py"
output="sine_plot" caption="Fourier partial sums
for n = 1, 3, 5, 9."}
```

3. DATA EXAMPLE

A simulated regression demonstrates inline code execution:

```
{python file=".inkwell/scripts/scatter.py"
output="scatter" caption="Bivariate scatter with
OLS regression."}
```

4. RESULTS

Table 1. Regression summary statistics.

Parameter	Value
n	150
r	0.894
Slope	0.700

5. CONCLUSION

The RMxAA template compiles from markdown with the journal's native two-column layout, dual-language abstracts, and standard bibliography formatting. As demonstrated in section 2 and section 3, Inkwell's code blocks produce embedded figures without manual export steps, following the literate programming paradigm ().

References

Knuth, Donald E. 1984. "Literate Programming." *The Computer Journal* 27 (2): 97–111. https://doi.org/10.1093/comjnl/27.2.97.

MacFarlane, John. 2023. *Pandoc: A Universal Document Converter*. Released. https://pandoc.org.